

Computer Architecture

Problem Set #1

Institute of Computer Science and Engineering
National Yang Ming Chiao Tung University

This problem set is **ungraded and not collected**. We will release solutions two weeks after the problem set is released. Please stop by office hours if you have questions.

1. Compute the Clocks Per Instruction (CPI) of a machine which has an average CPI for ALU operations of 1.1, a CPI for branches/jumps of 3.0, and a hit rate of 60% in the cache. A hit in the cache takes 1 cycle pipelined and a cache miss takes 120 cycles. Assume 22% of instructions are loads, 12% are stores, 20% are branches/jumps and the balance are ALU operations.

Solution:

$$\begin{aligned}\text{CPI} &= P_{\text{ALU}} \times \text{CPI}_{\text{ALU}} + P_{\text{BR/JMP}} \times \text{CPI}_{\text{BR/JMP}} + P_{\text{LD/ST}} \times (P_{\text{HIT}} \times \text{CPI}_{\text{HIT}} + P_{\text{MISS}} \times \text{CPI}_{\text{MISS}}) \\ &= (1 - 0.22 - 0.12 - 0.2)(1.1) + (0.2)(3) + (0.22 + 0.12) [(0.6)(1) + (1 - 0.6)(120)] \\ &= 17.63 \text{ CPI}\end{aligned}$$

2. You are a processor designer and have to make a decision between building a processor which executes at 1GHz and has an average CPI 1.2 and a processor which executes at 2GHz, but has a CPI of 2. Which is better to build and why?

Solution:

$$Performance = \frac{\text{Instructions}}{\text{Program}} \times \frac{\text{Cycles}}{\text{Instruction}} \times \frac{\text{Time}}{\text{Cycle}}$$

Case A: 1GHz, CPI = 1.2

$$Performance_A = k \times 1.2 \times 1 \text{ ns} = 1.2k \text{ ns}$$

Case B: 2GHz, CPI = 2

$$Performance_B = k \times 2 \times 0.5 \text{ ns} = 1k \text{ ns}$$

Assuming the same number of instructions in the program, k, we see that Case B has lower running time, therefore the 2GHz, CPI 2 machine is better.

3. For the following RISC-V code snippet, identify all of the RAW, WAW, and WAR hazards. Provide a list for each hazard. Hint: remember that you have to check more than just neighboring instructions.

```
add x1, x2, x3
sub x3, x4, x6
mul x5, x4, x7
addi x5, x5, 1
sub x6, x3, x9
andi x2, x1, x9
```

Solution:

- RAW

Register	Source	Destination
x1	add	andi
x3	sub1	sub2
x5	mul	addi

- WAW

Register	Source	Destination
x5	mul	addi

- WAR

Register	Source	Destination
x3	add	sub1
x6	sub1	sub2
x2	add	andi

4. For each of the characteristics below, complete the table by checking the appropriate entry (Architecture or Implementation) to indicate whether each feature is most often a characteristic of a computer architecture or of a particular CPU implementation.

Characteristic	Architecture?	Implementation?
# of bits in an instruction		
# of registers visible to programmers		
# of floating-point divider units		
RISC vs. CISC		
# of instructions issued per cycle		
Clock rate of the processor		
Pipeline depth		

Solution:

Characteristic	Architecture?	Implementation?
# of bits in an instruction	✓	
# of registers visible to programmers	✓	
# of floating-point divider units		✓
RISC vs. CISC	✓	
# of instructions issued per cycle		✓
Clock rate of the processor		✓
Pipeline depth		✓

5. Page B-60 in H&P6, Problem B.2 a,b

Solution:

a) **Fully associative cache**

Cache Block	Possible Memory Blocks
0	M0–M31
1	
2	
3	
4	
5	
6	
7	

b) **4-way set associative cache**

Cache Block	Set	Way	Possible Memory Blocks
0	0	0	M0, M2, M4, M6, M8, M10, M12, M14, M16, M18, M20, M22, M24, M26, M28, M30
1	0	1	
2	0	2	
3	0	3	
4	1	0	M1, M3, M5, M7, M9, M11, M13, M15, M17, M19, M21, M23, M25, M27, M29, M31
5	1	1	
6	1	2	
7	1	3	

6. How many SRAM bits are required to implement an 8KB two-way set-associative cache with a 64-byte block size? Assume that each cache line includes a single valid bit and excludes dirty bits. Additionally, consider there is one bit per set for true LRU implementation. Assume that the address size of the machine is 32-bits and that the machine allows for byte addressing.

Solution:

68,288 bits

$[(2 \text{ ways} * (20 \text{ tag bits} + 1 \text{ valid bit})) + 1 \text{ LRU bit}] * 64 \text{ lines}] + \text{data bits} = (2*21+1)*64+8*1024*8 = 68,288 \text{ bits}$

7. Using figure B.9 on page H&P6 B-25 and table B.8 on page B-24. Which has a lower miss rate, a 256KB direct mapped cache or a 64-KB 8-way cache? Which of the three C's drives the previous result?

Solution:

256KB direct mapped miss rate: 0.013

64KB 8-way miss rate: 0.029

256KB direct mapped has lower miss rate.

The bulk of the misses are caused by capacity which is why the direct mapped larger cache wins. If conflict dominated, larger cache could lose.